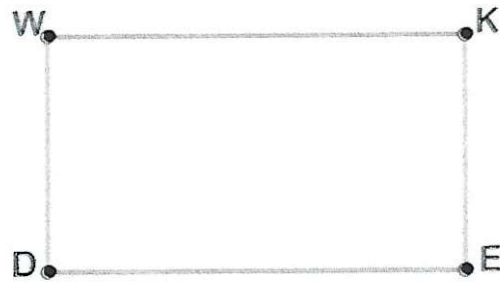
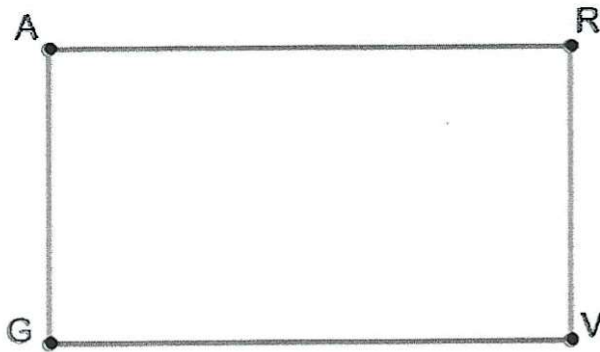


Unit 6: Worked Examples

1. Given $WD = 2$, $AR = (7w - 30)$, $RV = 2.5$, and $WK = (6w - 26)$. What is the value of w ?

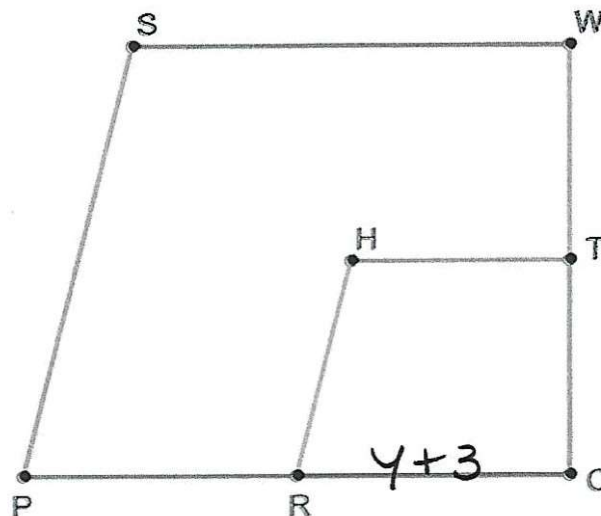


$$\frac{2}{2.5} = \frac{6w - 26}{7w - 30}$$

$$14w - 60 = 15w - 65$$

$$\boxed{5 = w}$$

2. Given $PC = (3y + 4)$, $RC = (y + 3)$, $SP = 8.2$, $SW = (7x - 13)$, $HT = (2x - 2)$, and $HR = 4.1$, what is the length of $\overline{RC}$?



$$\frac{4.1}{8.2} = \frac{y + 3}{3y + 4}$$

$$12.3y - 16.4 = 8.2y + 24.8$$

$$4.1y = 41$$

$$y = 10$$

$$\boxed{RC = 13}$$